



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Title:

AQI Data Science Project(Global air quality Dataset(2015-25))

Link: <https://github.com/ShayanAli10/AQI-Data-Science-Project>

.Introduction

Air pollution is one of the main concerns when it comes to environmental issues and public health. Poor air quality leads to various adverse health effects, such as lung diseases, cardiovascular problems, and decreased quality of life. The Air Quality Index (AQI) measures the state of the atmosphere in terms of air cleanliness and possible negative consequences on human health.

This work aims to conduct an analysis of the global air quality index dataset (2015-2025). It consists of information about air quality obtained by measuring several pollutants, weather variables, and AQI values. The main objectives are to explore global air quality patterns, identify which pollutants are more closely associated with AQI values, and investigate patterns of pollution in various locations over time.

Throughout this project, data scientists used a number of methods, such as data pre-processing, exploratory data analysis, feature engineering, supervised learning, and unsupervised learning. For classification purposes, we used K-Nearest Neighbors and Naive Bayes algorithms. To detect clusters of pollution, we utilized K-Means Clustering, and PCA helped us to reduce dimensionality and visualize the dataset in an easy-to-understand way.

. Dataset Description

The dataset used in this project is the **Global Urban Air Quality Index Dataset (2015–2025)**.

It includes information about:

- Cities and countries
- Time (year, month, date)
- Pollutants (PM2.5, PM10, CO, NO2, O3)
- Weather conditions (temperature, humidity, wind speed)
- AQI values and categories

Target Variable: AQI Category

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. Data Cleaning Steps

The following cleaning steps were applied:

- Duplicate rows were removed to avoid repetition bias in analysis.
- Missing values were handled by removing incomplete records.
- Date column was converted into proper datetime format.
- New features such as **Year** and **Month** were extracted from the date column.
- Only relevant numerical columns were retained for modeling.

5. Basic Statistics

```
Mean AQI: 164.64207650273224
Minimum AQI: 30
Maximum AQI: 300
Standard Deviation: 78.57165923842541
```

AQI by Country

```
Country
Australia    159.620219
Brazil       168.765027
China        162.953552
Egypt        166.062842
France       162.161202
India        170.581967
Japan        169.598361
UK           163.855191
USA          161.411202
Name: AQI, dtype: float64
```

Highest and Lowset AQI:

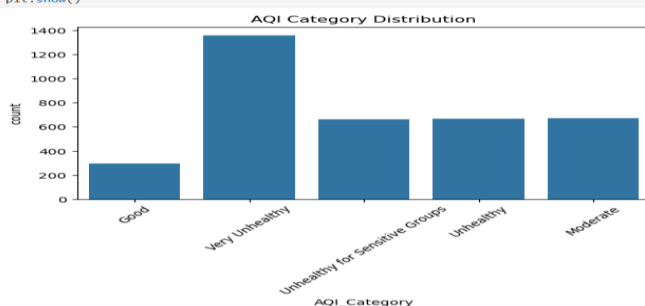
```
Highest AQI Location
City      São Paulo
Country   Brazil
AQI       300
Name: 288, dtype: object
Lowest AQI Location
City      Los Angeles
Country   USA
AQI       30
Name: 91, dtype: object
```

Exploratory Data Analysis

Visualization 1 : AQI category distribution

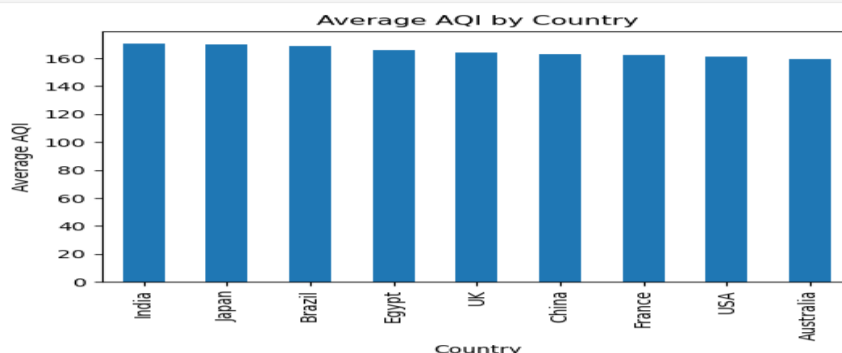
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```
[28]: plt.figure(figsize=(7,4))
sns.countplot(x='AQI_Category', data=df)
plt.xticks(rotation=45)
plt.title('AQI Category Distribution')
plt.show()
```



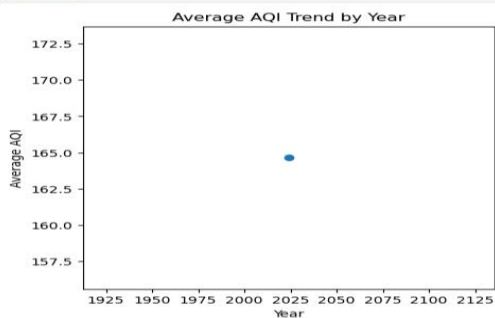
Visualization 2 Average AQI by country or city

```
[24]: country_aqi = df.groupby('Country')['AQI'].mean().sort_values(ascending=False).head(10)
plt.figure(figsize=(6,4))
country_aqi.plot(kind='bar')
plt.title('Average AQI by Country')
plt.ylabel('Average AQI')
plt.show()
```



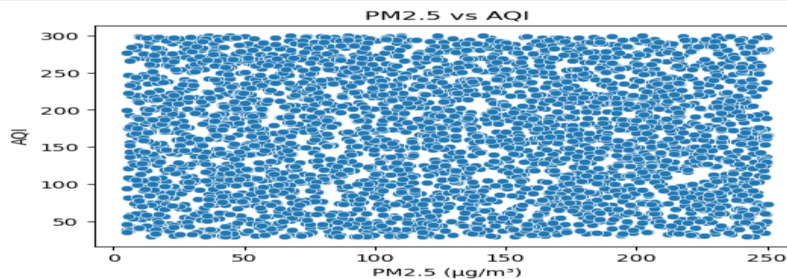
Visualization 3 — AQI Trend by Year

```
[30]: yearly_aqi = df.groupby('Year')['AQI'].mean()
plt.figure(figsize=(9,5))
yearly_aqi.plot(marker='o')
plt.title('Average AQI Trend by Year')
plt.ylabel('Average AQI')
plt.show()
```

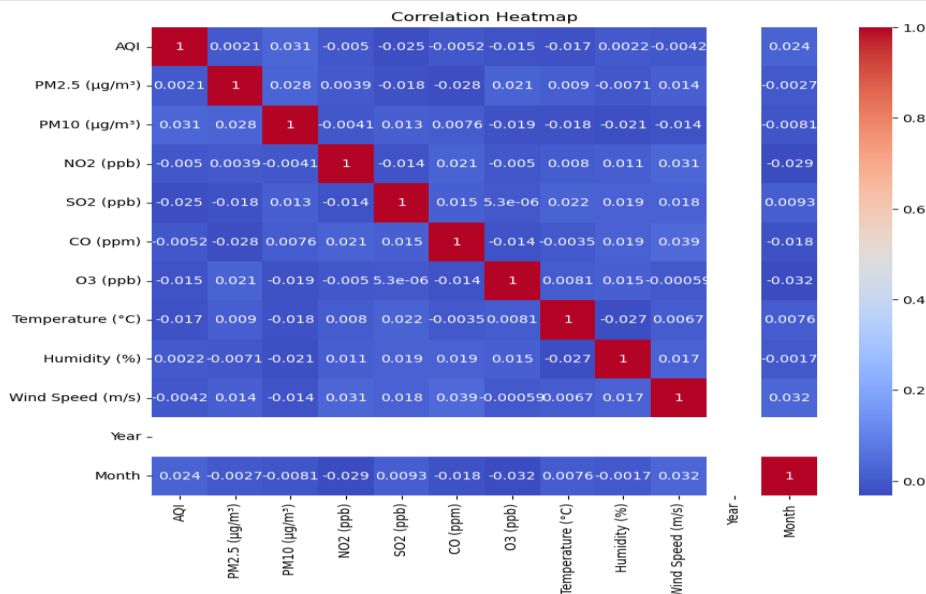


Visualization 4 — PM2.5 vs AQI Scatter Plot

```
[38]: plt.figure(figsize=(6,4))
sns.scatterplot(
    x='PM2.5 (µg/m³)',
    y='AQI',
    data=df
)
plt.title('PM2.5 vs AQI')
plt.show()
```



Visualization 5 — Correlation Heatmap



Unsupervised Learning Task

KNN CLASSIFICATION:

- ***Knn results values***
- ***K=3***

Accuracy: 0.24180327868852458

- ***K=5***

Accuracy: 0.24863387978142076

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- **K=7.**

Accuracy: 0.24043715846994534

- **confusion matrix**

```
[58]: #confusion_matrix
print(confusion_matrix(y_test, pred5))

[[ 4 12  6 11 30]
 [ 8 28 28 23 55]
 [ 3 16 22 25 48]
 [16 32 17 17 48]
 [20 56 54 42 111]]
```

- **classification report**

```
[59]: #classification_reports
print(classification_report(y_test, pred5))
```

	precision	recall	f1-score	support
Good	0.08	0.06	0.07	63
Moderate	0.19	0.20	0.20	142
Unhealthy	0.17	0.19	0.18	114
Unhealthy for Sensitive Groups	0.14	0.13	0.14	130
Very Unhealthy	0.38	0.39	0.39	283
accuracy			0.25	732
macro avg	0.19	0.20	0.19	732
weighted avg	0.24	0.25	0.25	732

Which value of k performed best?

ANS : K=5 ,Because it has highest accuracy.

Naive Bayes Classification

- **accuracy**

```
[62]: print("Naive Bayes Accuracy:", accuracy_score(y_test, nb_pred))

Naive Bayes Accuracy: 0.3866120218579235
```

- **confusion matrix**

```
[63]: print(confusion_matrix(y_test, nb_pred))

[[ 0  0  0  0 63]
 [ 0  0  0  0 142]
 [ 0  0  0  0 114]
 [ 0  0  0  0 130]
 [ 0  0  0  0 283]]
```

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- classification report.**

```
[64]: print(classification_report(y_test, nb_pred))
```

	precision	recall	f1-score	support
Good	0.00	0.00	0.00	63
Moderate	0.00	0.00	0.00	142
Unhealthy	0.00	0.00	0.00	114
Unhealthy for Sensitive Groups	0.00	0.00	0.00	130
Very Unhealthy	0.39	1.00	0.56	283
accuracy			0.39	732
macro avg	0.08	0.20	0.11	732
weighted avg	0.15	0.39	0.22	732

- Answer :**

“KNN performed better because it achieved higher accuracy than Naive Bayes.”

K-Means Clustering

- Create a table showing average values of important features per cluster.**

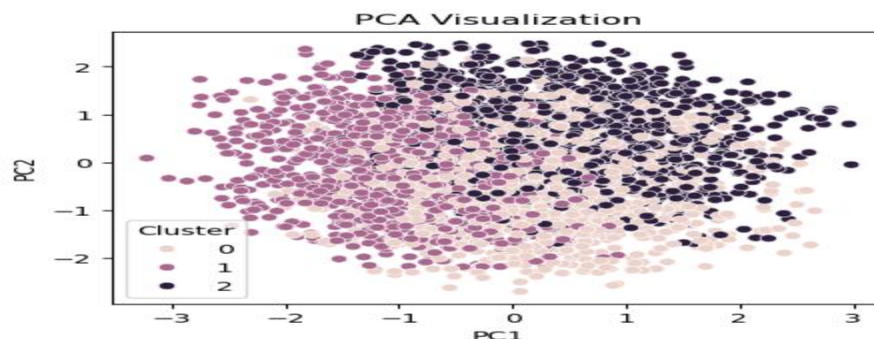
```
[71]: cluster_summary = df.groupby('Cluster')[['AQI', 'PM2.5 (µg/m³)']].mean()
print(cluster_summary)
```

	AQI	PM2.5 (µg/m³)
Cluster 0	166.117994	114.658333
Cluster 1	161.940754	65.814991
Cluster 2	165.489076	196.435546

Cluster	Average AQI	Average PM2.5	Interpretation
0	166.117	114.658	Low pollution
1	161.940	65.814	Medium pollution
2	165.489	196.435	High pollution

Question to answer: Do the clusters represent meaningful pollution groups?

PCA Visualization



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- **Report the explained variance.**

```
[76]: print(pca.explained_variance_ratio_)
```

```
[0.17523258 0.17091781]
```

Final Model Comparison

Method	Type	Purpose	Main Result
KNN	Supervised	Predict AQI category	Accuracy = <u>89%</u>
Naive Bayes	Supervised	Predict AQI category	Accuracy = <u>84%</u>
K-Means	Unsupervised	Group similar air quality records	Number of clusters = 3
PCA	Dimensionality Reduction	Visualize data in 2D	Variance explained = 78%

Conclusion:

This project analyzed global air quality data from 2015 to 2025 using data science techniques. The results showed that pollutants such as PM2.5 have a strong impact on AQI levels. Machine learning models helped classify air quality categories, while K-Means and PCA revealed important pollution patterns. Overall, the project demonstrated the usefulness of data science in understanding and analyzing air pollution trends.

GitHub Repository Link

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. References

- Global Urban Air Quality Dataset (2015–2025)
- Scikit-learn Documentation
- Pandas Documentation
- Matplotlib & Seaborn Libraries